Estimating Parameters in a Simple Linear Model.

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

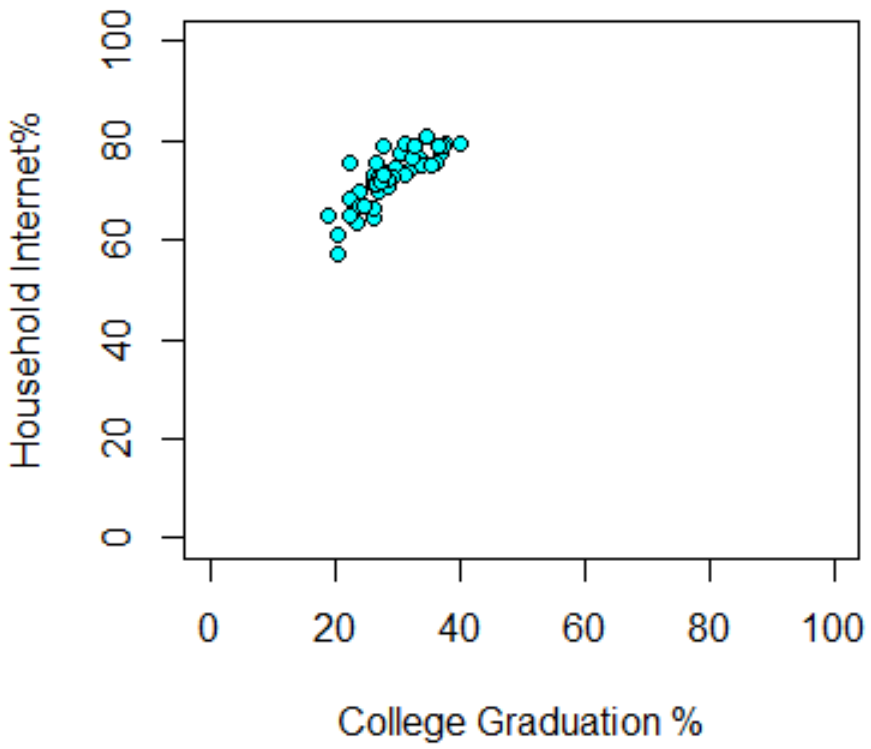
keith.lohse@health.utah.edu

@keith_lohse

HEALTH
UNIVERSITY OF UTAH

Internet Access Rates

```
> head(DAT1)
  State Internet College
1    AK     63.5    23.5
2    AL     79.0    28.0
3    AR     60.9    20.6
4    AZ     73.9    27.4
5    CA     77.9    31.0
6    CO     79.4    37.8
…
```



Model C:

$$y_i = \beta_0$$

Model A:

$$y_i = \beta_0 + \beta_1(x_i)$$

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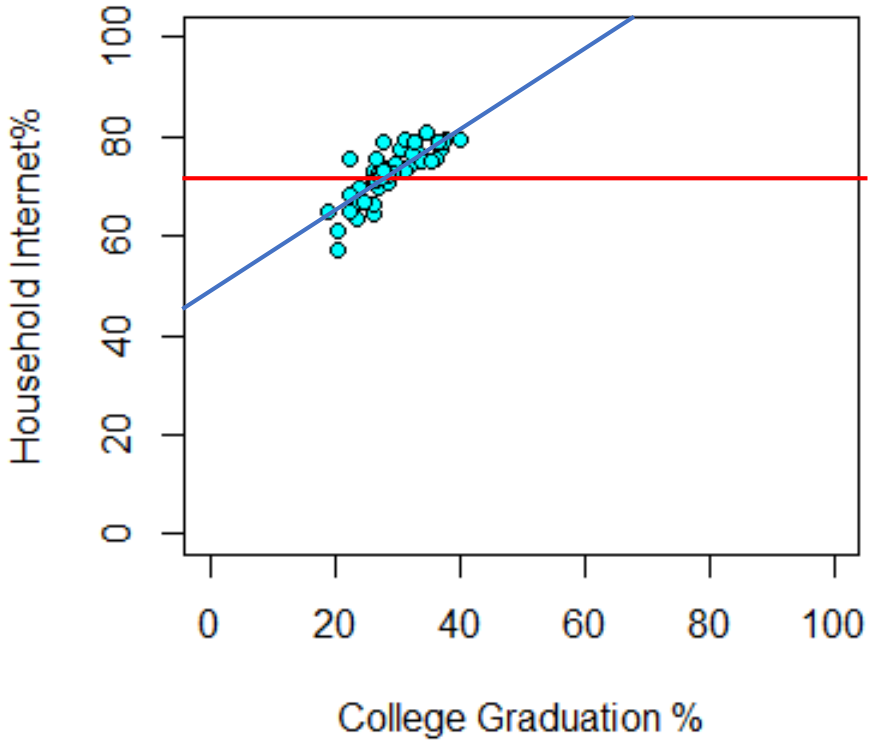
Which produces the smaller sum of squared errors?



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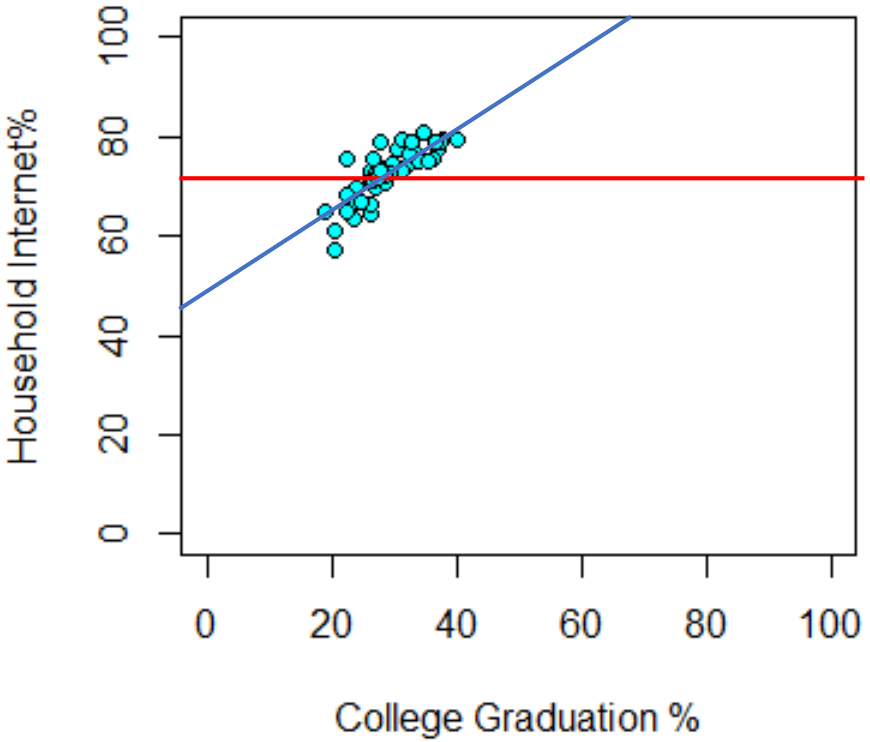

Model C:

$y_i = \beta_0$

Model A:

$y_i = \beta_0 + \beta_1(x_i)$

First, we need to be able to estimate the intercept and the slope!

Estimating Two Linear Parameters

Slope

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

Intercept:

$$b_0 = \bar{y} - b_1\bar{x}$$

```
DAT1$x_res<-DAT1$College-mean(DAT1$College)
DAT1$y_res<-DAT1$Internet-mean(DAT1$Internet)

DAT1$xy_cov<-DAT1$x_res*DAT1$y_res
DAT1$x_var<-DAT1$x_res^2

> head(DAT1)
 State Internet College  x_res   y_res    xy_cov     x_var
1    AK     63.5    23.5 -5.384  -9.306 50.103504 28.987456
2    AL     79.0    28.0 -0.884   6.194 -5.475496  0.781456
3    AR     60.9    20.6 -8.284 -11.906 98.629304 68.624656
4    AZ     73.9    27.4 -1.484   1.094 -1.623496  2.202256
5    CA     77.9    31.0  2.116   5.094 10.778904  4.477456
6    CO     79.4    37.8  8.916   6.594 58.792104 79.495056
```

Estimating Two Linear Parameters

Slope

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = 0.86$$

Intercept:

$$b_0 = \bar{y} - b_1\bar{x}$$

```
DAT1$x_res<-DAT1$College-mean(DAT1$College)
DAT1$y_res<-DAT1$Internet-mean(DAT1$Internet)

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> sum(DAT1$xy_cov)/sum(DAT1$x_var)
[1] 0.8618517
```

Estimating Two Linear Parameters

Slope

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = 0.86$$

Intercept:

$$b_0 = \bar{y} - b_1\bar{x} = 47.91$$

```
> mean(DAT1$Internet)-0.8618517*mean(DAT1$College)
[1] 47.91228
```

Derivation of the Slope and Intercept Estimates

- In the book, there is an algebraic derivation for these "least squares" estimates of the slope and the intercept.
  - Work through this derivation to make sure you understand it.

- Conceptually, however, lets focus on one question, why must this line always run through $\bar{y}$ and $\bar{x}$?
  - As a single value, the mean will always produce the smallest SSE.
  - If there was no relationship between x and y (i.e., $b_1 = 0$), the best model of y would be $\bar{y}$ (and the same for x and $\bar{x}$).
  - Thus, we see how x and y covary around their means $\sum(x_i - \bar{x})(y_i - \bar{y})$. As x departs from its mean, so does y (i.e., $b_1 \neq 0$), but the line 'pivots' at the mean of both variables.